

Lecture 2: ArcOnline

Designing basemaps for ArcCollector

Arc Collector

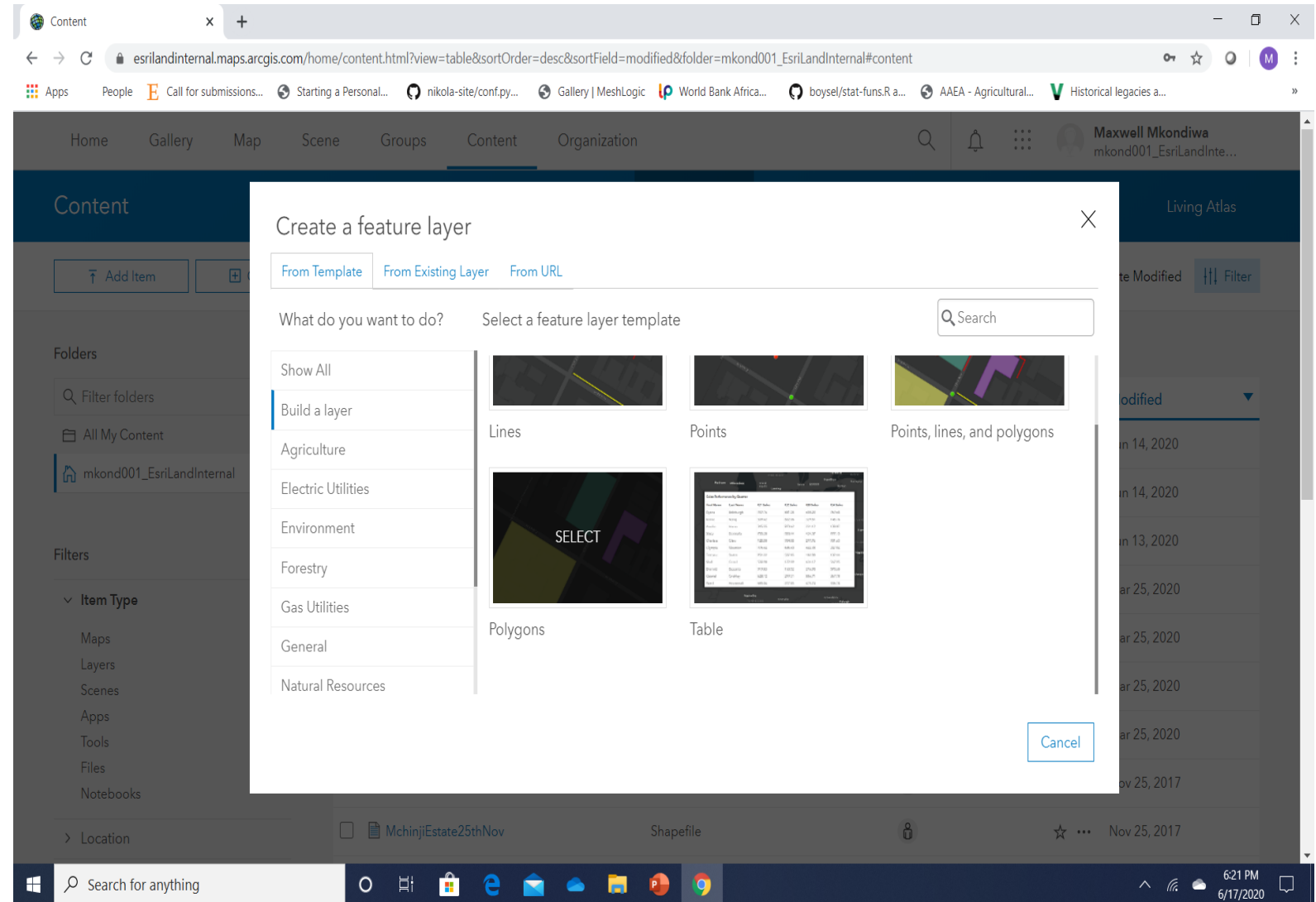
- Arc GIS Collector is an ESRI application (app) that is used to collect geo-mapping information
- Why ArcGIS Collector
 - It is faster.
 - It is easy to monitor progress.
 - One can easily add identifiers (and pictures) to the polygon drawn.
 - Uses the latest satellite imagery such that most features (e.g. roads, houses) can be seen on the app.

Steps

1. Create new feature class
2. Add the variables for the form
3. Add feature class to new map
4. Edit the style for the map to style you want for the mapping
5. Save and share the map to a group
6. Download the basemap for the shared map in ArcCollector for offline mapping

1. Create new feature class

- Select content
- Click create

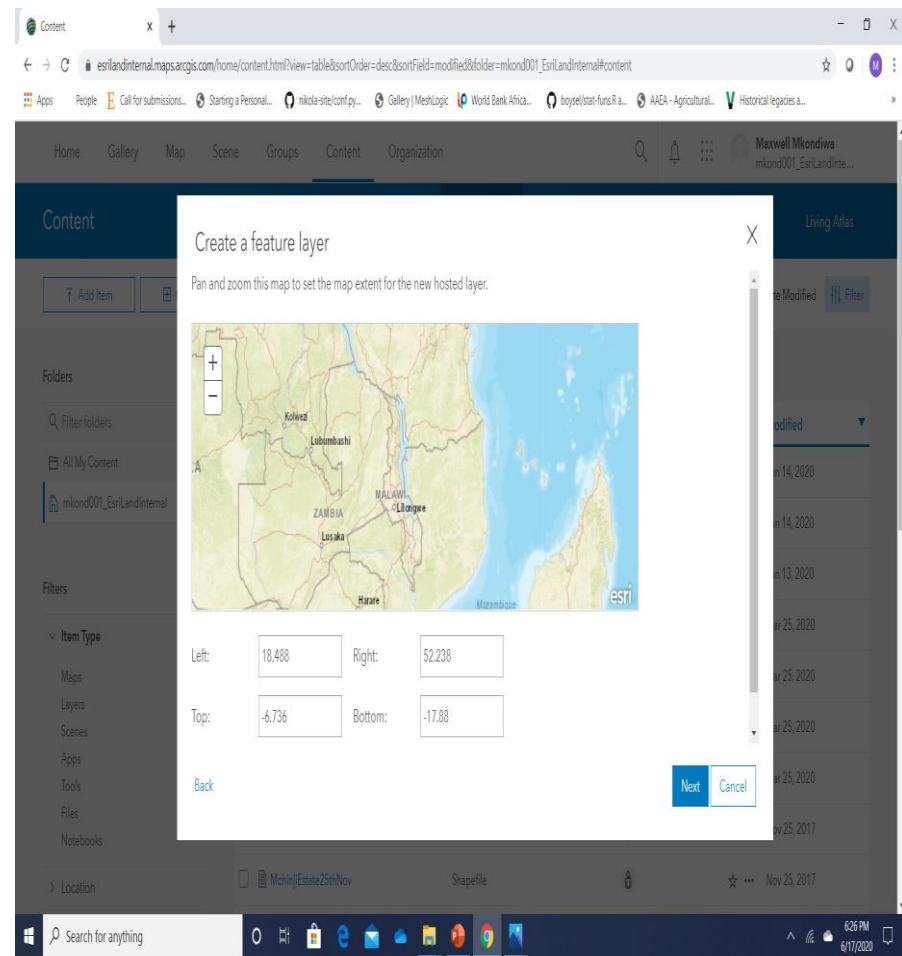
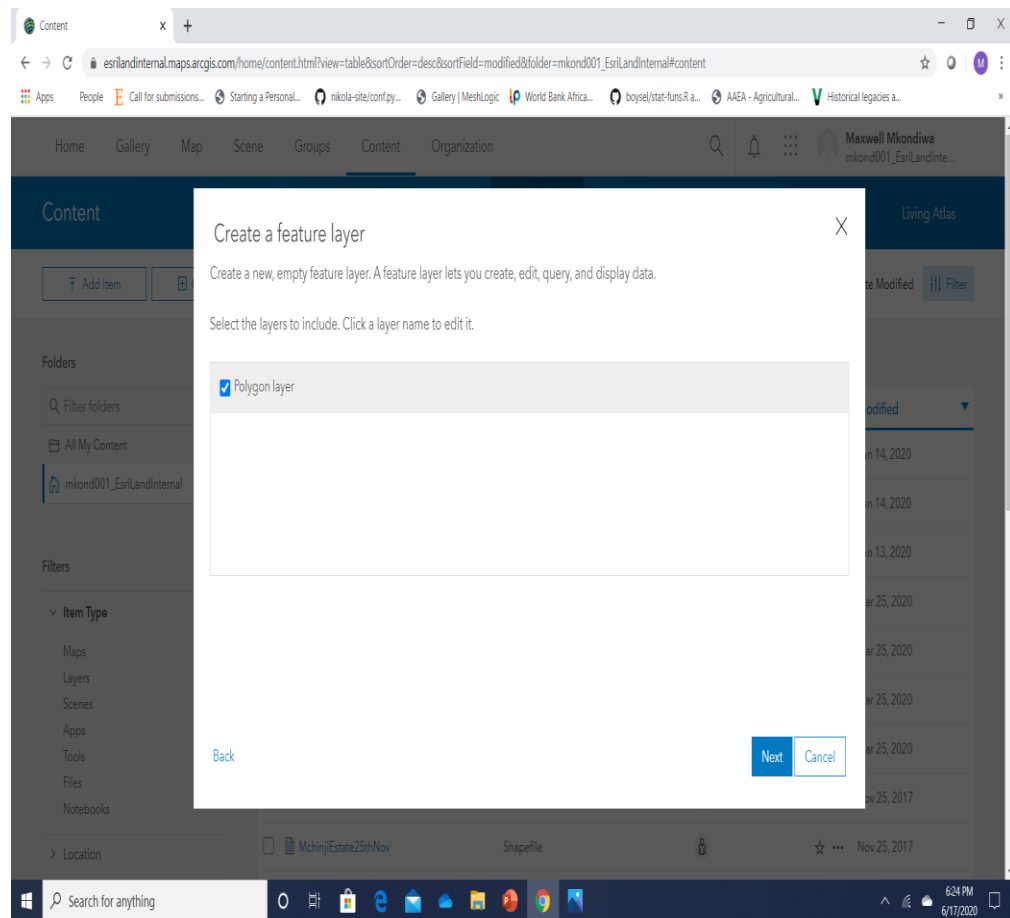


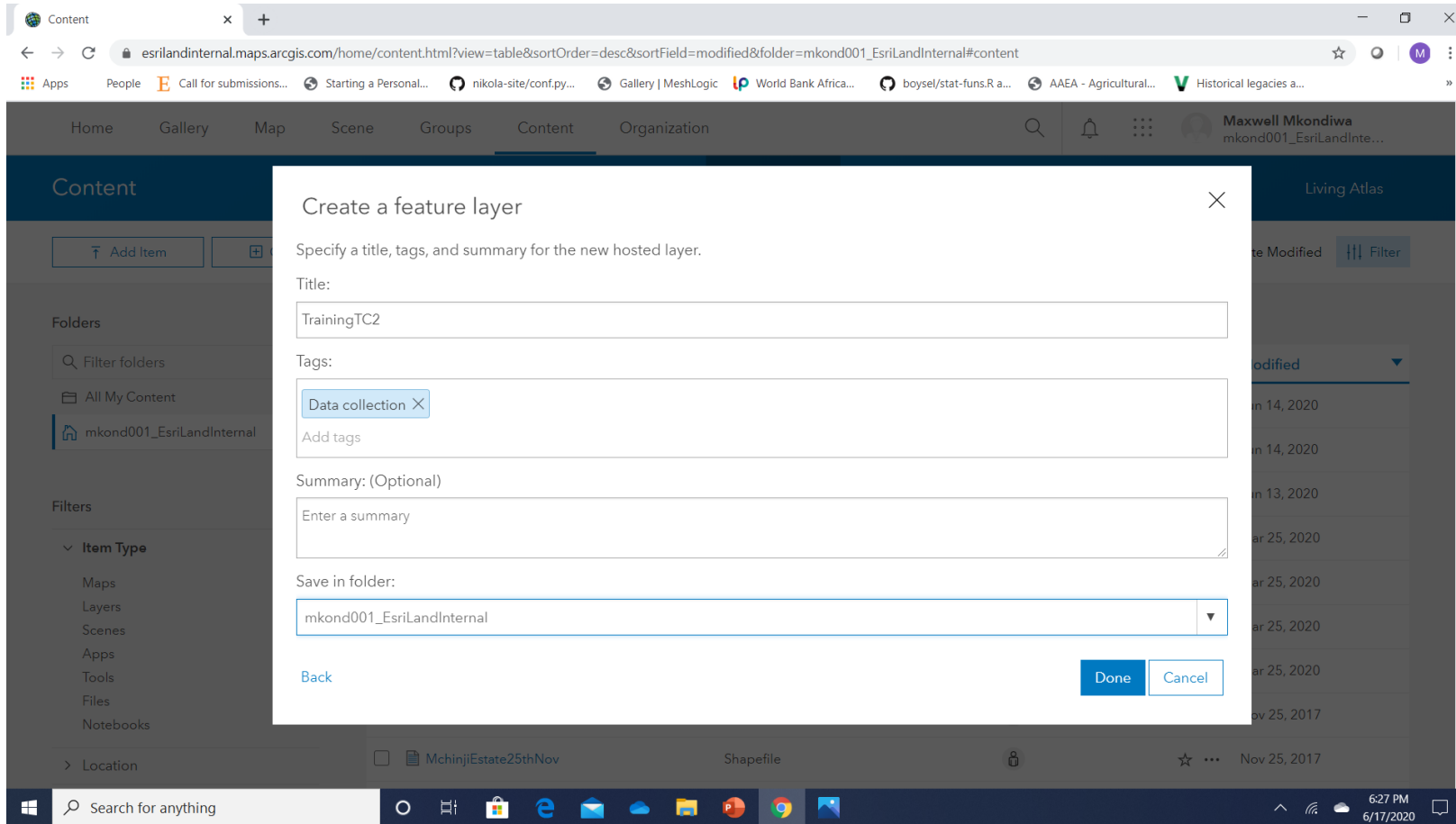
- Title
- Tags: data
- Summary

The screenshot shows the 'Create a feature layer' dialog box in the ArcGIS Online interface. The dialog is titled 'Create a feature layer' and has a close button (X) in the top right corner. It contains the following fields and options:

- Title:** A text input field containing 'TrainingTC'.
- Tags:** A section with a tag 'Data collection X' and an 'Add tags' button.
- Summary: (Optional)** A text input field containing 'TrainingTC'.
- Save in folder:** A dropdown menu showing 'mkond001_EsrilandInternal'.
- Buttons:** 'Back', 'Done', and 'Cancel' buttons at the bottom.

The background shows the ArcGIS Online 'Content' page with a sidebar containing 'Folders' and 'Filters' sections. The user's profile 'Maxwell Mkondiwa' is visible in the top right corner.





Windows taskbar showing search bar and various application icons (File Explorer, Edge, Mail, OneDrive, Photos, PowerPoint, Chrome, and Teams).

2. Data form

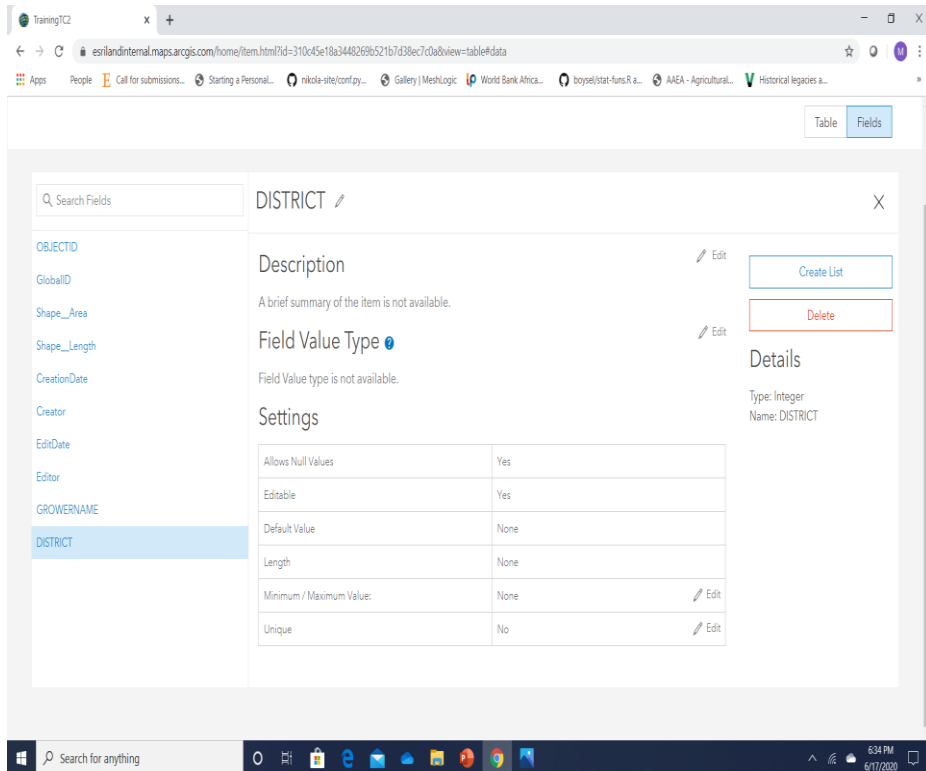
The screenshot shows the ArcGIS TrainingTC2 web application interface. The browser address bar displays the URL: `esrilandinternal.maps.arcgis.com/home/item.html?id=310c45e18a3448269b521b7d38ec7c0a&view=table#data`. The user is logged in as Maxwell Mkondiwa. The interface includes a top navigation bar with tabs for Home, Gallery, Map, Scene, Groups, Content, and Organization. A secondary navigation bar shows Overview, Data, Visualization, Usage, and Settings. The 'Data' tab is active, displaying a table view of the data. A modal dialog titled 'Add Field' is open, allowing the user to create a new field. The dialog contains the following fields and options:

- Field Name:
- Display Name:
- Type: - Length:
- Default Value: (Optional)
- Allow Null Values: ☒

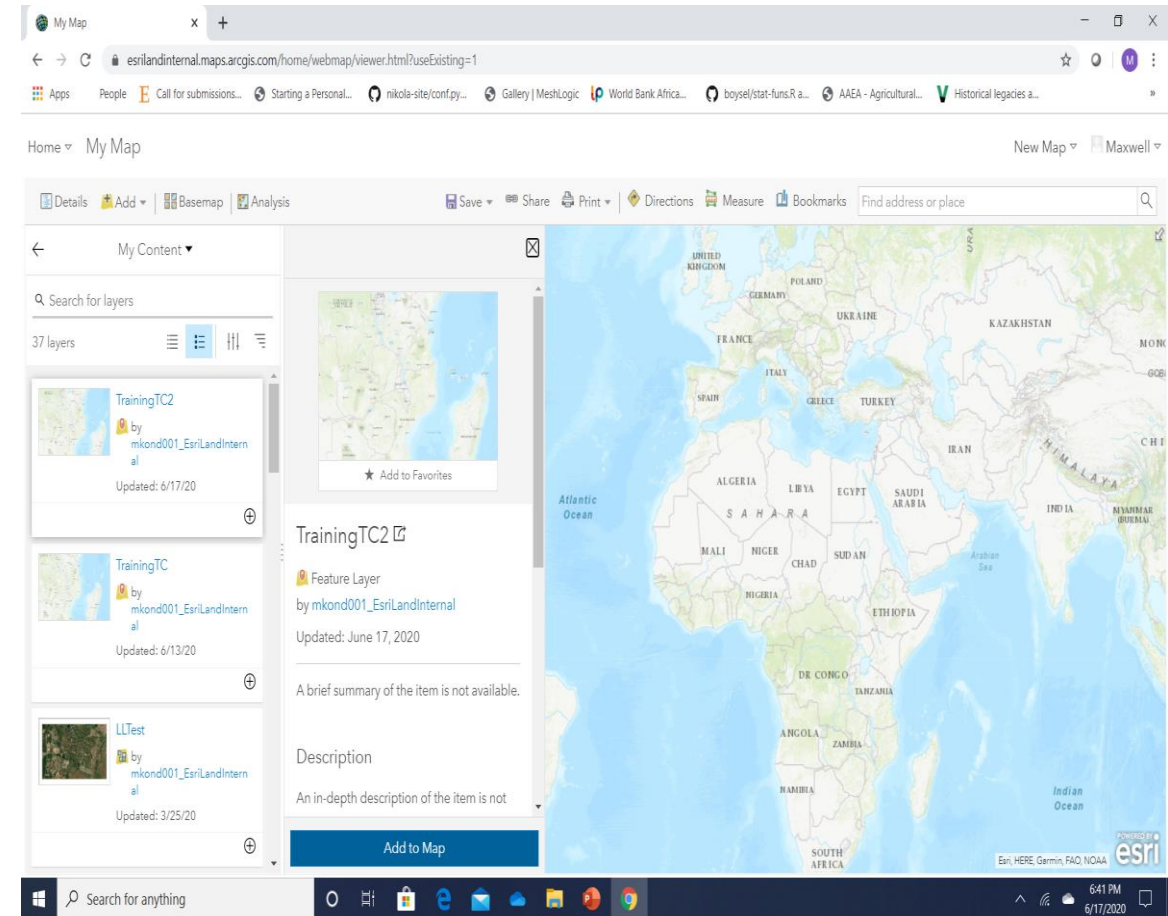
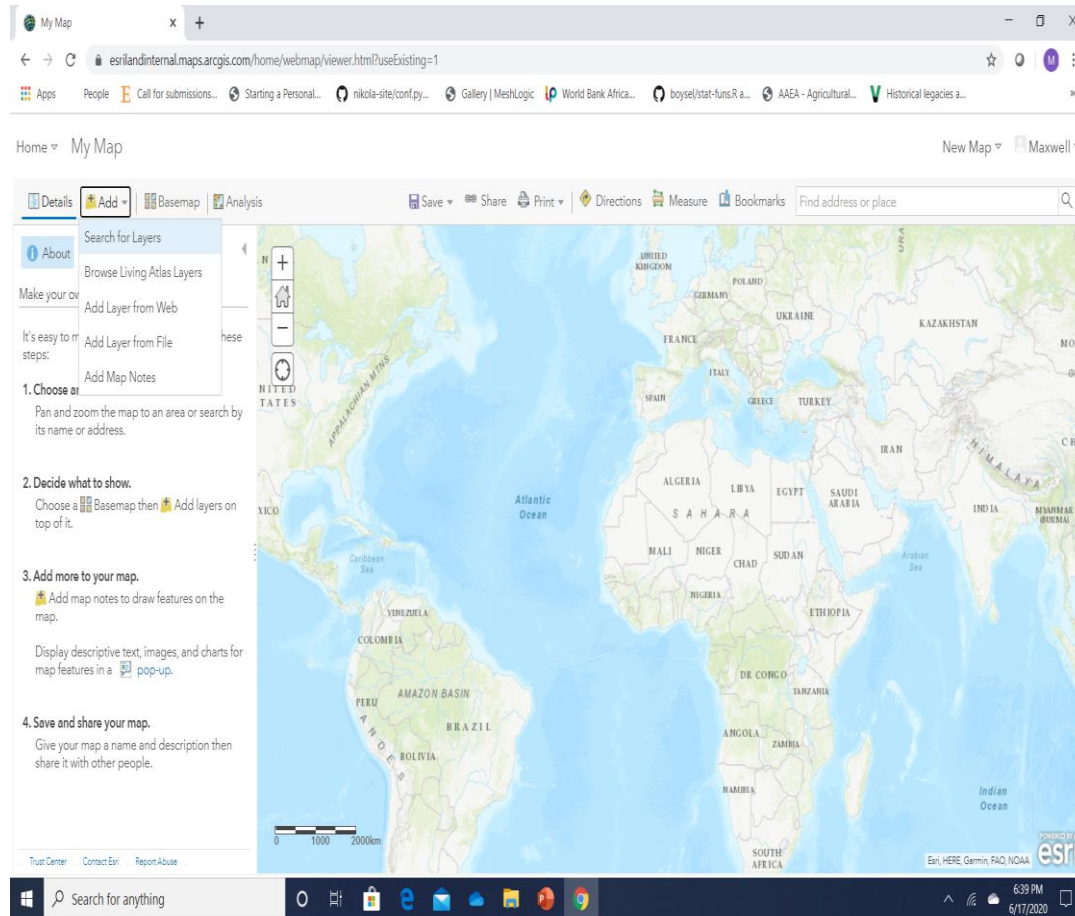
At the bottom of the dialog are two buttons: 'Add New Field' and 'Cancel'. The background interface shows a list of fields with checkboxes for selection and a 'Search Fields' input. The Windows taskbar at the bottom indicates the time is 6:32 PM on 6/17/2020.

Categorical variable with a list

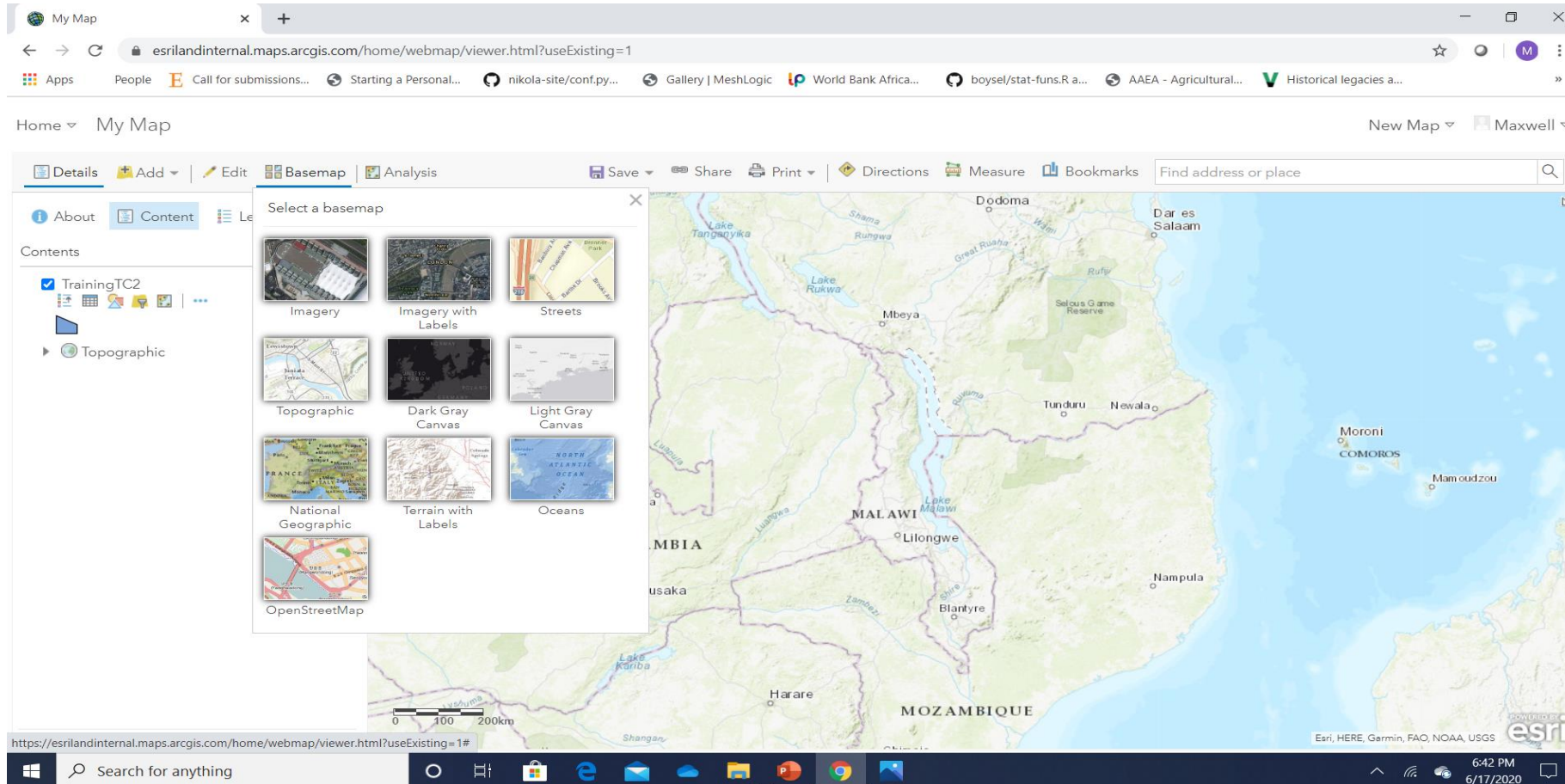
- For example: District
- Click on the variable then create list



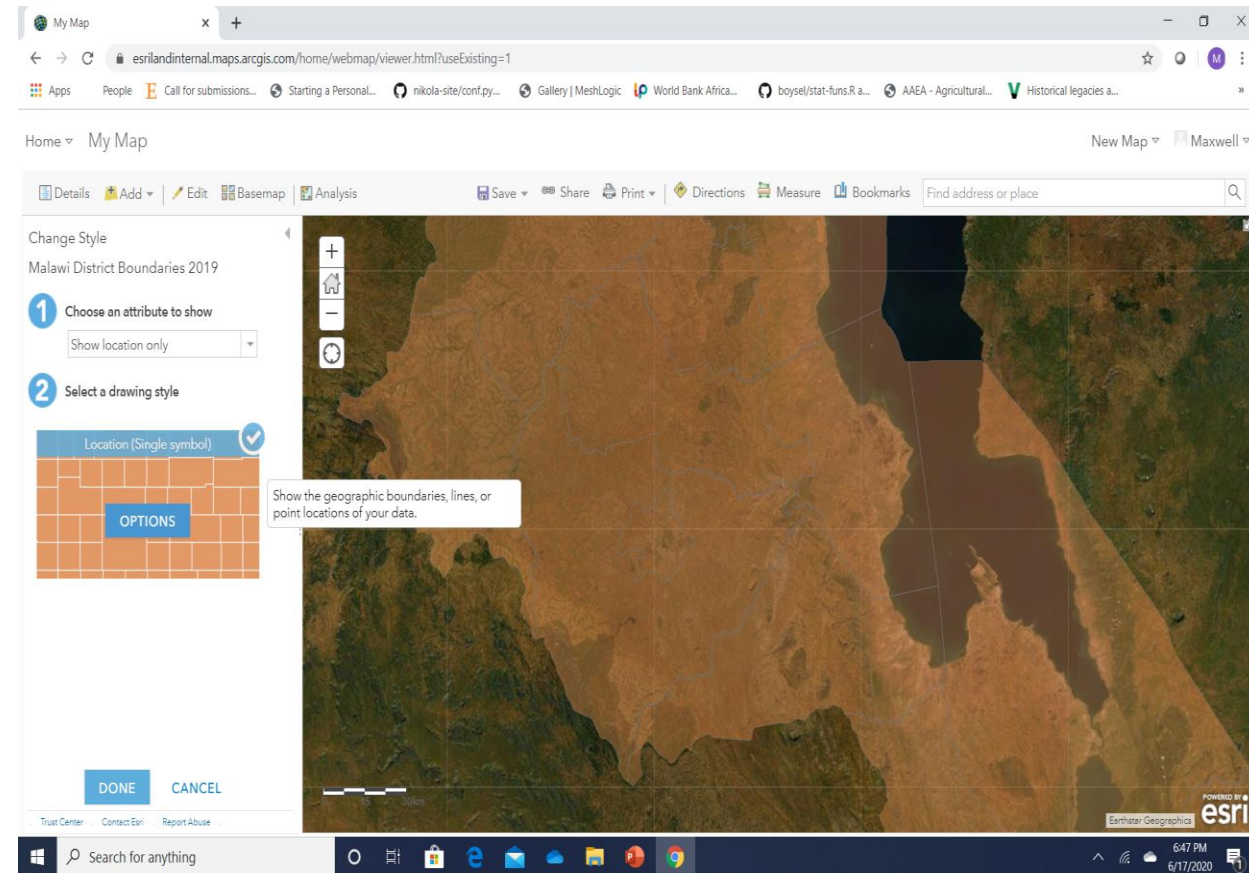
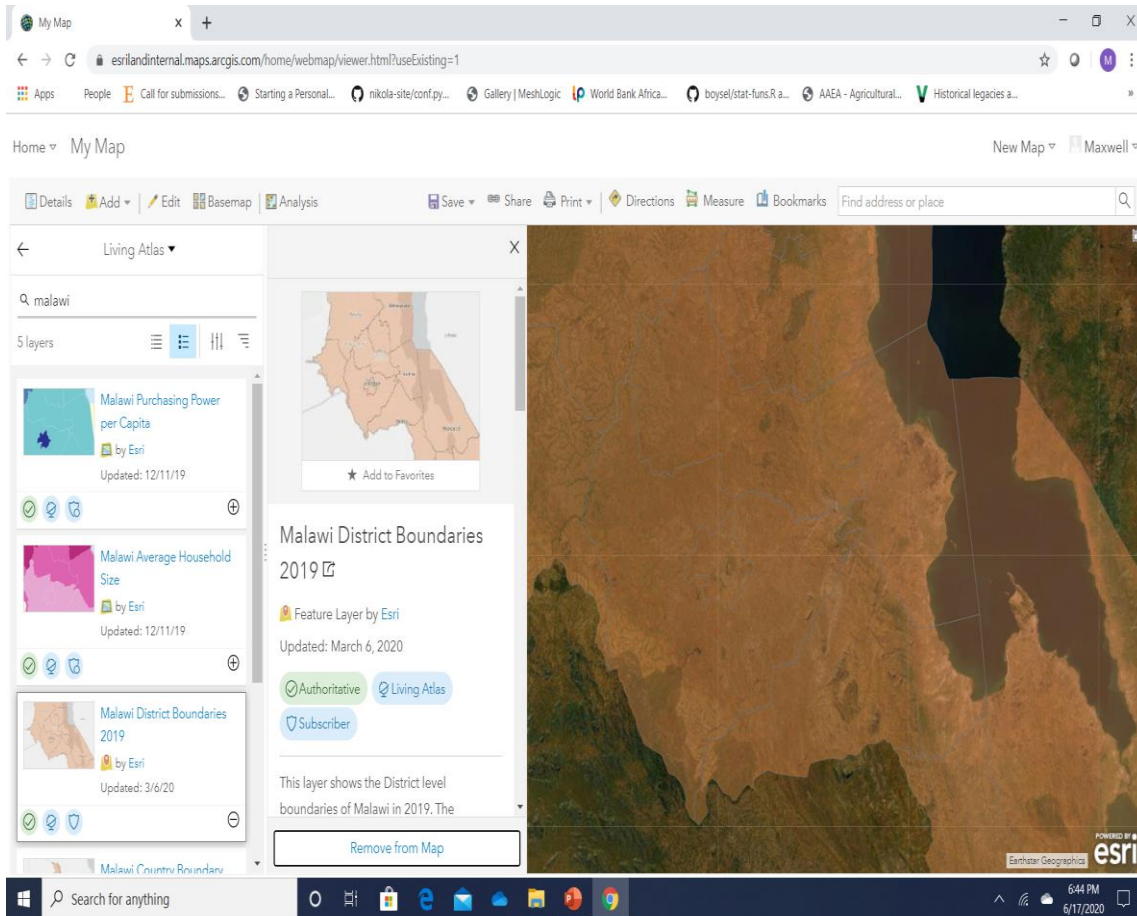
3. Add feature to a map



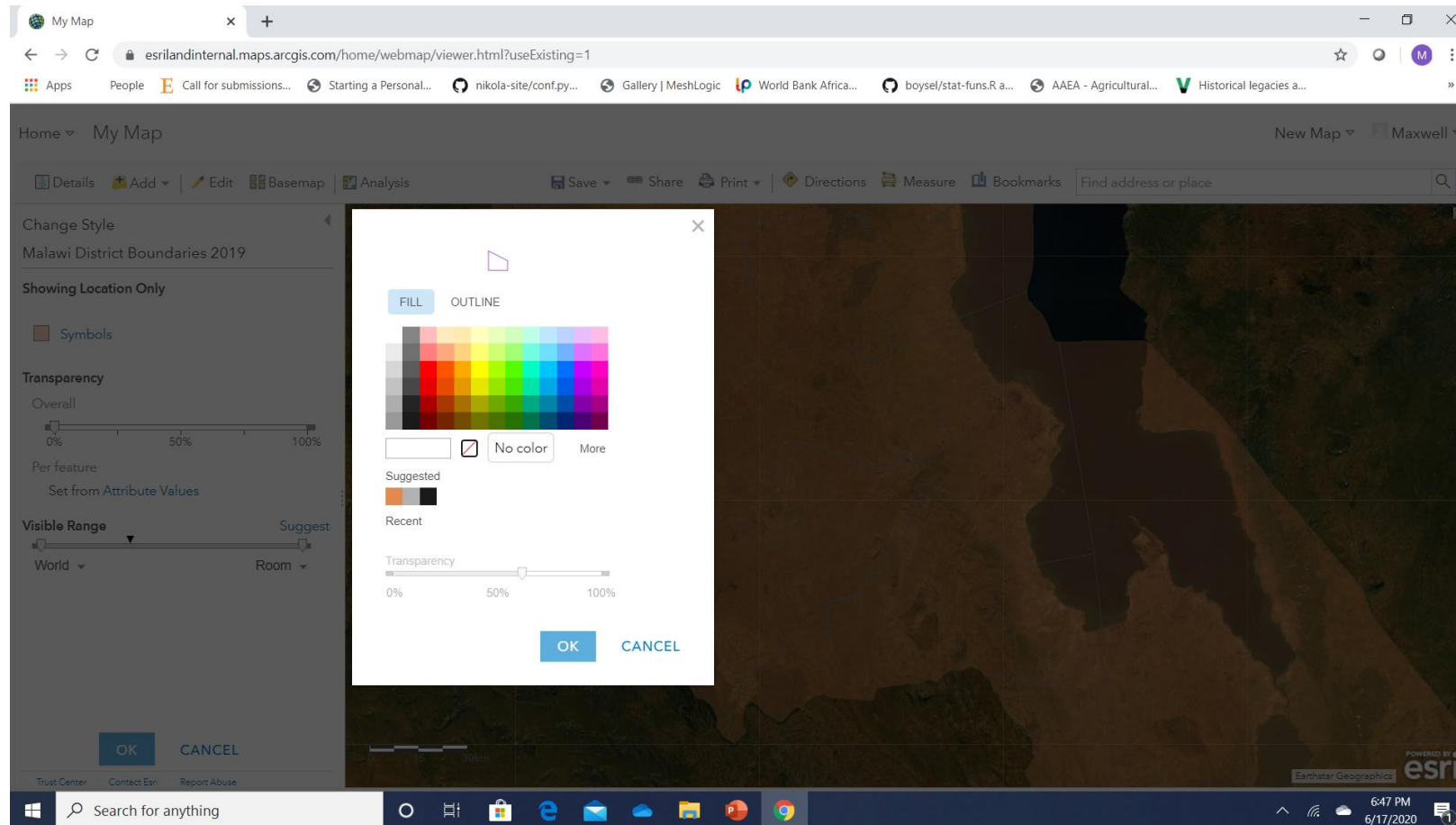
Change the basemap to imagery



4. Add boundaries and change style to outline so you can still see the imagery



Fill: no color and change outline width



If doing basemap for Lilongwe, filter

The screenshot shows the ArcGIS web map viewer interface. A filter dialog box titled "Filter: Malawi District Boundaries 2019" is open. The dialog has a "Create" tab and a text input field containing "Lilongwe". Below the input field, there are radio buttons for "Value", "Field", and "Unique", with "Value" selected. There are also buttons for "APPLY FILTER", "APPLY FILTER AND ZOOM TO", and "CLOSE". The background map shows Malawi with district boundaries. A table at the bottom of the screen displays data for Malawi districts.

ID	2019 Total Population	ISO_CODE	ISO_SUB	ISO2_CC	ISO3_CC
0101	886,702	MW		MW	MWI
0102	886,702	MW		MW	MWI
0103	886,702	MW		MW	MWI
0104	1,040,664	MW		MW	MWI
0105	633,565	MW		MW	MWI

The screenshot shows the ArcGIS web map viewer interface. The map displays Malawi with district boundaries highlighted in purple. A table at the bottom of the screen displays data for Malawi districts.

ID	2019 Total Population	ISO_CODE	ISO_SUB	ISO2_CC	ISO3_CC
0104	1,040,664	MW		MW	MWI
0110	1,722,574	MWLI	LI	MW	MWI

5. Save and share the map

- Once saved and shared, you can open in Arc Collector in the tablet

Using ArcGIS Desktop (not covered)

- The process of designing base maps and forms to be used for offline data collection in ArcGIS collector involves four main steps;
- Prepare and share forms for entering text data,
- Prepare and share tile packages for district, traditional authorities and enumeration area boundaries,
- Share and public feature service definition, tile packages and maps, and
- Download base maps for offline editing.

- Create a geodatabase form in excel
 - The form has the following important columns; input table, field name, field type, field precision, field scale, field length and field alias.
- In ArcGIS Desktop
 - Add the folder you are working in to the Arc Catalog. Right click on the folder, then select new>File Geodatabase.
 - Rename the geodatabase to the name as in the excel file. We then need to create the drop-down menus for the variables that have a coded list e.g. enumerator name, crop name, district.
 - To do this, right click on the name then click properties -> domains. Then enter the information for each of the coded lists.

- ***Assign file name***
- Right click on the geodatabase then Select Feature Class. Use WGS 1984 Mercator
- Enter a name like the ... \name in the excel file.
- ***Define geodatabase domains***
 - We need to fill in the other information. To do that open the Arc Toolbox in the Geoprocessing menu. Then select Data management tools-> Fields -> Add Field-> (Right click) Batch.
 - Insert as many rows as in the excel file. For now, highlight 7 rows. Copy all the contents in the excel file from 1 to 7 starting from the Input Table column up to the Field Domain column.
 - To avoid errors, make sure the file name in the Input Table column is the same as the name you provided to the geodatabase in Arc Desktop.

- ***Edit the styling of the shape in the geodatabase***
- ***Share as a feature service***
- Click on file and make sure you are signed in.
- The proceed on share as -> service-> Publish as service
- Select the following options:
 - Capabilities
 - Feature class
 - Feature access
 - Select all
- Feature class: Fill
- Item description: Fill
- Sharing: Share to the groups
- Then click publish. If you want pop ups of the shape to appear and be editable, make sure the shapefiles for the boundaries are open when sharing the service.

Using Tile Packages (not covered)

ArcGIS Desktop

- Select the shapefiles of the boundaries you want represented on the map then select **shares as-> tile packages->** on the home tab.
- Make sure the tile package created is at 20 of 20 levels, that is, at building level. While the imagery is what matters for mapping, the boundaries allow the research team to navigate from one village to another and therefore efficiently identify households. In a country where transport costs are high, this can help in reducing survey costs and improving the quality of the data collected.
- *Add layers to appear in the collector using Arc Map Online*
- In Arc Online, click on contents. Then share the service definition for the service you created.
- Select: Map-> search for layers-> add the new layer-> share it
- The tile packages and the feature service should be added to a new map.

Using Tile Packages (not covered)

- ArcGIS Desktop Model Builder
 - Copy and paste the toolbox into the folder you are using for the project.
Open a new model, save it with a name of your choice.
- Tile Package Kreator
- The tile package kreator is used to upload- more or less manually- each enumeration area boundary/ traditional authority boundary into tablets. Open the tile package Kreator
 - Sign in with your Esri username and password
 - Click create tile package (second and middle option)
 - Scroll the projection to 17 (Number of zoom levels)
 - Insert the name of the tiles
 - Drag the json files on at a time into the Kreator
 - Click create tile package.